OSF Amendment 1 – Pre-registration of Calibrated Federated Crash and Near-Crash Risk Forecasting on SHRP2

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1. Status of this amendment

This amendment adds one sensitivity analysis and replaces one decision rule in the registration filed on OSF Registries on 4 October 2026 (https://osf.io/y5zrp/overview). Every hypothesis, primary endpoint and test in the confirmatory family stays as registered.

Data access. At the date of this amendment the author has not accessed the SHRP2 bird′s-eye trajectory dataset in any version. The Data Use License application to VTTI is still pending. No SHRP2 count, summary or result has informed any part of this amendment.

Basis. Both changes come from a full dress rehearsal of the registered analysis on the development data (INTERACTION, stand-in outcome), together with simulations whose true answers are known. Section 6 reports the rehearsal results; they are development evidence only and do not count as confirmatory evidence for any hypothesis.

2. Reason for the amendment

Before requesting confirmatory data, the author rehearsed the complete registered analysis on INTERACTION. The rehearsal ran every confirmatory model with the five registered seeds and the registered training budget (20 pretraining and 10 fine-tuning epochs), then applied the registered tests, bootstrap and Holm correction. To mirror the structure of the SHRP2 driving domains, it used four binary domain factors with unequal level sizes. The factor rules were committed to version control before any model was trained.

The rehearsal exposed two design weaknesses in the registered H2 analysis. Neither can be detected without data; both can be corrected now, before the confirmatory data exist.

1. **Domain shift is confounded with training-set size.** In the registered H2 comparison, each domain level′s model is trained only on that level′s data. Small levels (e.g. adverse weather, wet surface, darkness) therefore yield models trained on far less data. A model can be worse calibrated simply because it saw less data, so the H2 ratio partly measures data quantity rather than domain shift.

2. **The H2 secondary criterion treats correlated frames as independent.** Predictions are made at 10 Hz, so consecutive frames from one trip are nearly identical. The registered Clopper–Pearson interval counts every frame as an independent trial. Its interval is therefore far too narrow, and violations of the false-alarm guarantee are declared too often.

3. Change 1: size-matched H2 as a pre-specified sensitivity analysis

The registered H2 test and its role as the primary endpoint stay unchanged. A size-matched version is added and reported next to it for every factor-direction pair; it is not part of the Holm family.

Definition. For each domain factor, the level with more hazard-training samples is called the larger level.

- Its training units (drivers, or events if no driver identifier exists) are put in random order using the training seed of the model concerned.
- Whole units are added in that order until the subset's hazard-training sample count reaches or first exceeds that of the smaller level.
- A model is pretrained and fine-tuned on this subset only, with the same settings, seeds and budget as every other model. Model selection uses the larger level's full validation data, unchanged.
- The size-matched H2 statistic is the registered mean log ratio of adaptive ECE at 3 s, with the size-matched model replacing the full larger-level model. The replacement applies in both directions: as the model trained elsewhere when the smaller level is held out, and as the in-domain reference when the larger level is held out. Testability rules, units and bootstrap resamples are those of H2.

Interpretation. The size-matched result serves as a robustness check on H2.

- If H2 is supported and the size-matched estimate has the same sign with a 95% interval excluding zero, the conclusion is reported as robust to training-set size.
- If the two disagree, the disagreement is reported, and the claim is limited accordingly.

Rationale. In the rehearsal, models trained on the size-matched subset had higher validation loss than full models of the same level. The gap grew with the size ratio:

Size ratio	Higher validation loss
1.5×	+4%
2.8×	+10%
3.2×	+12%

Size ratio	Higher validation loss
4.4×	+14%

This confirms that training-set size alone moves calibration and loss.

4. Change 2: cluster-bootstrap interval for the H2 secondary criterion

Registered rule (replaced). "A held-out domain violates the conformal guarantee if the lower bound of the 95% Clopper–Pearson interval of its false-alarm rate exceeds α."

New rule. A held-out domain violates the conformal guarantee if the lower bound of the 95% percentile interval of its false-alarm rate, from the paired cluster bootstrap, exceeds α. The bootstrap uses the same units (drivers, or events) and the same B = 2,000 resamples (seed 131) as all confirmatory tests.

- The false-alarm rate is computed on the held-out level's test negatives, with the conformal threshold fitted on the source level's validation negatives, exactly as registered.
- The number of violating domains is reported for α = 0.05 and 0.01, as registered.
- The Clopper–Pearson interval is still reported for transparency, but it no longer decides.

Rationale. The registered interval assumes independent trials, but consecutive 10 Hz frames are strongly dependent. The author tested both intervals in a simulation with known truth: 200 replicates, 40 clusters of 100 strongly correlated frames each, and a conformal threshold at α = 0.05. A valid 95% interval should place its lower bound above the true false-alarm rate in about 2.5% of replicates.

- **Clopper–Pearson:** 32.0% of replicates.
- **Cluster bootstrap:** 0.5% of replicates.

In the rehearsal, Clopper–Pearson flagged 26 of 35 held-out pair × seed combinations as violations at α = 0.05, against 11 of 35 for the cluster interval.

5. What stays unchanged, and clarifications

Everything not named in Sections 3 and 4 remains exactly as registered, including:

- the four hypotheses and their directions;
- the nine confirmatory tests and the Holm family;
- endpoints, horizons and protocols P1 and P2;

- domain factors and client definitions, and the small-domain rules;
- models, frozen settings, seeds 131–135 and the training budget;
- the bootstrap, minimum effects of interest, and all sensitivity and exploratory analyses.

Neighbour augmentation stays frozen. In the rehearsal the augmented model did not meet the H4 clean non-inferiority margin (−0.022 AUPRC against a margin of −0.02). The settings are deliberately not retuned on these results. If H4 fails on SHRP2, the failure will be reported.

Clarification of labels (no change of substance). Section 6 of the registration uses sub-labels that Section 1 does not define. They mean:

- **H2a:** the H2 primary test (mean log ECE ratio).
- **H2b:** the H2 secondary criterion (false-alarm guarantee).
- **H3b:** its two tests, FedAvg vs local and FedPer vs local.
- **H4a, H4b, H4c:** the three H4 tests in the order listed: natural radar dropout, 50% synthetic flicker, and clean non-inferiority.

Feasibility. The registered training budget is feasible on the author's encrypted CPU-only computer, where the licensed data must stay. The rehearsal of all H1, H2 and H4 models (70 models) took 29.1 hours, and re-fitting ten models reproduced their predictions exactly.

Planned amendment 2. As stated in the registration, the Phase 2 calibration method will be registered in a separate amendment before it is evaluated on SHRP2 test data.

6. Development evidence from the rehearsal (not confirmatory)

The registered tests were rehearsed on INTERACTION, with the stand-in outcome (moving-pair TTC2D < 1.5 s within 3 s) and protocol P2.

- **Test set:** 16,805 test samples, 320 positives, 22 test recordings as bootstrap units, B = 2,000.
- **Seeds:** five for H1, H2 and H4; one for H3.
- **Stand-ins:** LightGBM replaced GSSM in H1a. The H4a subset is a weak proxy on drone data, where a missing neighbour frame usually means a vehicle entering the scene rather than a sensor dropout.

Test	Estimate [95% CI]	p (Holm)	Result
H1a model − LightGBM, AUPRC	+0.145 [0.065, 0.210]	0.0045	supported

Test	Estimate [95% CI]	p (Holm)	Result
H1b model − TTC2D (moving), AUPRC	+0.171 [0.084, 0.247]	0.0045	supported
H2 mean log ECE ratio	+0.378 [0.206, 0.470]	0.0045	supported
H3a ECE FedAvg − FedPer	+0.0063 [−0.0002, 0.0100]	0.124	not supported
H3b FedAvg − local, AUPRC	+0.011 [−0.047, 0.081]	1.000	not supported
H3b FedPer − local, AUPRC	+0.076 [0.019, 0.124]	0.0125	supported
H4a natural-dropout subset	−0.028 [−0.050, −0.005]	1.000	not supported
H4b 50% flicker	+0.154 [0.095, 0.212]	0.0045	supported
H4c clean non-inferiority (margin −0.02)	−0.022 [−0.033, −0.011]	1.000	not supported

H2 robustness:

H2 version	Mean log ratio [95% CI]	Out-of-domain ECE
Registered	+0.378 [0.206, 0.470]	1.46×
Size-matched	+0.343 [0.158, 0.488]	1.41×
After temperature recalibration	+0.222 [0.075, 0.391]	1.25×

Seven of eight factor-direction pairs were testable.

Records. Code, factor rules and analysis scripts were committed to version control before each corresponding run. The commit history, logs and result files will be released with the paper.

Use of AI assistance. Code development and drafting were assisted by an AI system (Claude, Anthropic), as disclosed in the registration. All analyses were run, checked and approved by the author.